Skills Summary

Scatter Plots and Bivariate Association

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a scatter plot and its trend

Read the plot in this order:

- **Direction** — dots rising left to right is a positive association; falling is negative; no slant is no association.
- **Strength** — tightly packed around a line is strong; widely scattered is weak.
- **Outliers** — one dot far off the pattern of the rest. Note it before you summarise anything, because it moves means and it weakens the trend.
- **Range** — the smallest and largest x-values in the data. Write them down: every prediction gets judged against them.

Example: A plot has the points $(1, 12)$, $(2, 15)$, $(3, 26)$, $(4, 21)$, $(5, 24)$. Which x-value carries the highest y-value, and does that point fit the pattern?
Step 1. Highest y means scan up the vertical axis first and only then read down to the horizontal axis — do not assume the largest x carries it.
Step 2. The top point is 26, sitting above $x = 3$, while its neighbours at $x = 2$ and $x = 4$ are only 15 and 21. The rest of the plot climbs gently, so this dot sits well above the pattern: it is an outlier. $\Rightarrow$ $x = \mathbf{3}$, and it does *not* fit the trend

Try it: A plot has the points $(2, 40)$, $(4, 44)$, $(6, 68)$, $(8, 55)$, $(10, 60)$. Which x-value carries the highest y-value?

check: $x = \mathbf{6}$

Skill 2 — Predicting from a trend line

Rule: a trend line $y = mx + b$ turns any input into a prediction. Substitute the x-value and evaluate — the slope m *multiplies*, it never adds.

Slope in words: m is the predicted change in y for each one-unit increase in x. **Intercept in words:** b is the prediction at $x = 0$, which is often outside the data and therefore often meaningless.

Example: A trend line is $y = 1.5x + 20$. Predict y when $x = 12$.
Step 1. A prediction from a line is just the line evaluated at that input — substitute and compute, keeping the slope as a multiplier.
Step 2. $y = 1.5(12) + 20 = 38$, and in words: each extra unit of x is worth about 1.5 more y. $\Rightarrow$ $y = \mathbf{38}$

Try it: A trend line is $y = 2.5x + 10$. Predict y when $x = 14$.

check: $y = \mathbf{45}$

Skill 3 — Qualifying a prediction

Every prediction needs a sentence about how far to trust it. Check all four:

- **Inside the range?** Predicting inside the observed x-values is *interpolation* (supported). Outside it is *extrapolation* (assumed).
- **Possible?** An answer past a real limit — a score above 100, a negative length — shows the line has been pushed past where it can hold.
- **Outlier-driven?** Re-summarise without the odd point and see how far the answer moves.
- **Cause or company?** An association is not evidence that one variable *makes* the other change.

Residual = actual − predicted. Positive means the line under-predicted that point; negative means it over-predicted.

Example: The line is $y = 1.5x + 20$. A real data point is $(20, 47)$. Find the residual and say what it tells you.
Step 1. Residual is actual minus predicted, in that order, because the sign is the whole message — so compute the prediction and subtract it from the observed value.
Step 2. $47 - (1.5(20) + 20) = -3$, and the negative sign says the observed value sits below the line. ⇒ $\mathbf{-3}$, so the line *over*-predicted this point by three

Try it: The line is $y = 2x + 12$. A real data point is $(25, 66)$. Find the residual.

check: **4**

(!) Watch out: the arithmetic of a prediction is never the hard part — a line returns a number for any input, including inputs no data covers and outputs the situation cannot produce. Before writing a prediction down, say out loud which observed range it sits in. And never upgrade “these two move together” into “this one causes that one”.